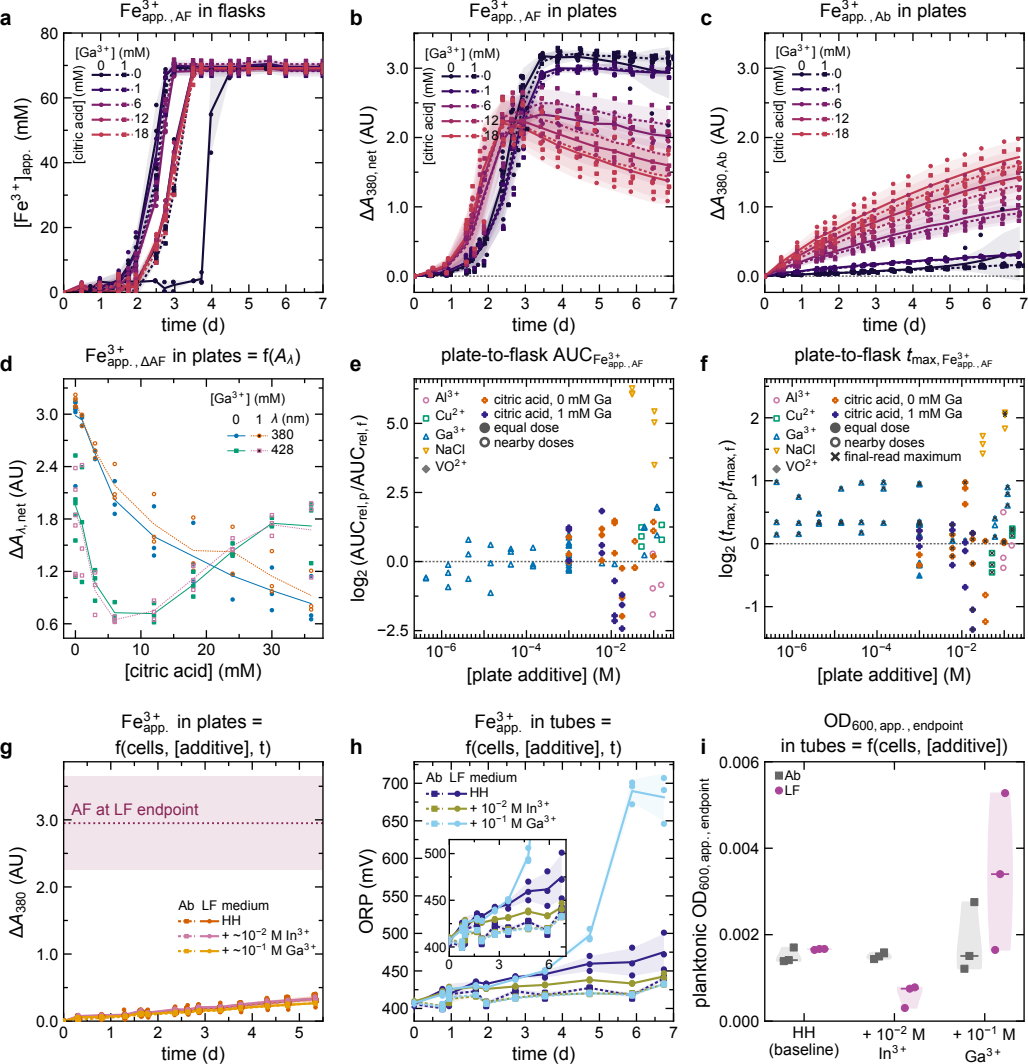

a
$Fe^{3+}_{app.,AF}$ in flasks
$[Fe^{3+}]_{app.}$ (mM)
time (d)
[Ga^{3+}] (mM)
0 1 ·0 ·1 ·6 ·12 ·18
[citric acid] (mM)
b
$Fe^{3+}_{app.,AF}$ in plates
$\Delta A_{380,net}$ (AU)
time (d)
c
$Fe^{3+}_{app.,Ab}$ in plates
$\Delta A_{380,Ab}$ (AU)
time (d)
d
$Fe^{3+}_{app.,\Delta AF}$ in plates = f(A_λ)
$\Delta A_{\lambda,net}$ (AU)
[citric acid] (mM)
[Ga^{3+}] (mM)
0 1 λ (nm)
380
428
e
plate-to-flask $AUC_{Fe^{3+}_{app.,AF}}$
$\log_2$ ($AUC_{rel,p}$/$AUC_{rel,f}$)
[plate additive] (M)
Al^{3+}
Cu^{2+}
Ga^{3+}
NaCl
VO^{2+}
citric acid, 0 mM Ga
citric acid, 1 mM Ga
equal dose
nearby doses
f
plate-to-flask $t_{max,Fe^{3+}_{app.,AF}}$
$\log_2$ ($t_{max,p}$/$t_{max,f}$)
[plate additive] (M)
final-read maximum
g
$Fe^{3+}_{app.}$ in plates = f(cells, [additive], t)
ΔA_{380} (AU)
time (d)
AF at LF endpoint
Ab LF medium
HH
+ ~10^{-2} M In^{3+}
+ ~10^{-1} M Ga^{3+}
h
$Fe^{3+}_{app.}$ in tubes = f(cells, [additive], t)
ORP (mV)
time (d)
Ab LF medium
HH
+ 10^{-2} M In^{3+}
+ 10^{-1} M Ga^{3+}
i
$OD_{600,app.,endpoint}$ in tubes = f(cells, [additive])
planktonic $OD_{600,app.,endpoint}$
Ab
LF
HH (baseline)
+ 10^{-2} M In^{3+}
+ 10^{-1} M Ga^{3+}